

Linear constraints

Vertices and active constraints

Michel Bierlaire

Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

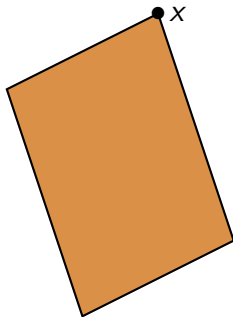
Vertices

The vertices of a polyhedron play a major role in optimization. Often, this is where we will find the optimal solution.

Vertex

x is a vertex of $\mathcal{P}$ if there is no $y, z \in \mathcal{P}$ such that $\exists 0 < \lambda < 1$ and

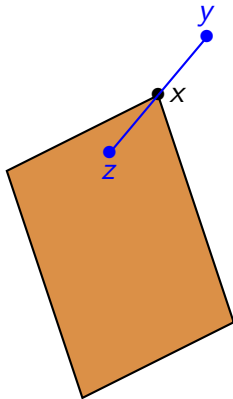
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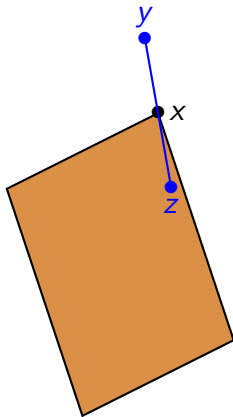
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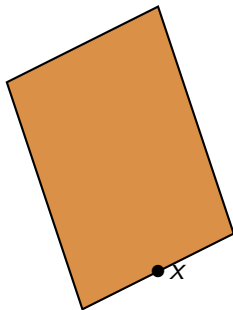
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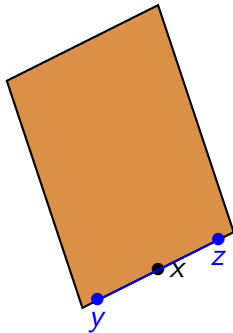
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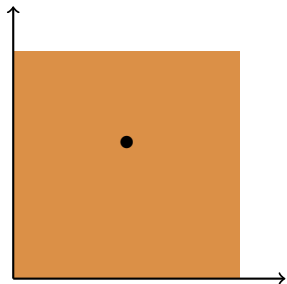
Existence

Theorem 3.37

- ▶ Consider a polyhedron $\mathcal{P}$ in standard form.
- ▶ If it is not empty, it has at least one vertex.

Idea of the proof

- ▶ Start from $x \in \mathcal{P}$.
- ▶ Follow a direction pointing to a constraint.
- ▶ Activate the first constraint met.
- ▶ Repeat in the facet which is of lower dimension.



Algebraic representation

Intuition

- At a vertex, constraints are active.

$$\begin{aligned} AP &= (B|N) \\ x &= P \begin{pmatrix} B^{-1}(b - Nx_N) \\ x_N \end{pmatrix} = P \begin{pmatrix} B^{-1}b \\ 0_{\mathbb{R}^{n-m}} \end{pmatrix}. \\ B^{-1}b &\geq 0. \end{aligned}$$

Theorem 3.35

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Algebraic representation

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- ▶ At a vertex, constraints are active.
- ▶ Standard form: $x_i = 0$.
- ▶ To find a vertex:
 1. Select and eliminate basic variables.
 2. Set all non basic variables to 0.
 3. Check feasibility.

$$AP = (B|N)$$
$$x = P \begin{pmatrix} B^{-1}(b - Nx_N) \\ x_N \end{pmatrix} = P \begin{pmatrix} B^{-1}b \\ 0_{\mathbb{R}^{n-m}} \end{pmatrix}.$$
$$B^{-1}b \geq 0.$$

Theorem 3.35

Summary

Vertices play an important role in optimization

We have provided a geometrical and an algebraic representations

Finding a vertex amounts to find the constraints that must be activated

In standard form, it means choosing a basis, set the non basic variables to zero, and check feasibility.